

Lejun Min

Audio ML Research Engineer

- contact@lejun.site
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- LinkedIn
- GitHub
- Mandarin / English

Profile

Audio ML researcher at Stanford CCRMA working on generative audio and music, representation learning, and audio signal processing. Experience spans Qwen Audio, Smule, and Sony CSL; publications include ICLR (Spotlight) and ISMIR.

Technical Skills

Audio ML

- Audio & Music Generation
- Audio Tokenization / Neural Codecs
- Audio Representation Learning
- Audio Captioning
- Music Information Retrieval
- Audio Deepfake Detection
- Perceptual Audio Coding
- Digital Signal Processing

ML Methods

- Diffusion Models
- Transformers
- Self-Supervised Learning
- Multimodal Learning
- Contrastive Learning

Languages & Frameworks

- Python
- C++
- Rust
- PyTorch
- TensorFlow
- Hugging Face
- torchaudio
- librosa
- Accelerate
- Lightning

Audio Tools

- JUCE
- Chuck
- Ambisonics

ML Systems

- Linux
- Docker
- Distributed Training
- SLURM
- MLflow

Education

Stanford University, CCRMA

2024 – Expected Jan. 2027

M.A. in Music, Science, and Technology (GPA 4.0/4.0)

- Advisors: Prof. Marina Bosi, Prof. Takako Fujioka
- CCRMA Flagship Project Award, Chiang Chen Overseas Graduate Fellowship

Shanghai Jiao Tong University, Zhiyuan College

2019 – 2023

B.Eng. in Computer Science (Fellowship)

- Member of **ACM Honor Class** – elite CS program for top 5% students
- Top 1% Outstanding Bachelor's Thesis (41 out of 3873 graduates)

Experience

Qwen Audio, Tongyi Lab, Alibaba Group – Research Intern

June 2026 – Present

- Co-developed Qwen-Audio-3.0-Gen-Preview, a unified generative model for long-form, multi-character audio scenes with explicit temporal control.
- Improved downstream semantic alignment of an RVQ-based tokenizer by 25% using contrastive learning and multi-stage training.
- Contributed to curation and post-processing of 1M+ hours of audio data.

Smule Labs – Research Intern

Oct. 2025 – Dec. 2025, Mar. 2026 – May 2026

- Developed an end-to-end audio-captioning framework for disentangled music-semantic representations; achieved >0.4 Spearman audio-caption alignment in internal evaluations and validated specialization across five musical attributes.

Sony CSL Paris – Research Intern

June 2025 – Sept. 2025

- Built a two-stage generative music remixing system using mix-invariant representations and conditional generation.
- Reduced the mix-to-content variation ratio by >90% in internal ablations, without requiring stems at inference.

SoundPatrol & Stanford – Research Assistant

Feb. 2025 – May 2025

- Trained singer-identity representations for singing-voice deepfake detection; reduced EER by 10 percentage points and reached 96% top-1 accuracy on SingFake. The system has been integrated into SoundPatrol's infringement detection pipeline.

Music X Lab, MBZUAI – Research Assistant

June 2022 – Dec. 2022, Sept. 2023 – Feb. 2024

- Developed a four-stage cascaded diffusion system over a hierarchical symbolic language, generating structured pop songs up to 256 measures and significantly outperforming baselines on long-range structure (ICLR 2024 Spotlight, top 5%).
- Developed Polyffusion, a piano-roll DDPM supporting five generation and control tasks via masked inpainting and cross-attention; achieved the best objective results on 4/5 tasks and significant gains in naturalness and musicality (ISMIR 2023).

Publications

Qwen-Audio-3.0-Gen-Preview Technical Report

arXiv preprint

J. Dai, ... L. Min, et al. (equal contribution, alphabetical order)

Leveraging Rotational M/S Coding and ML in Stereo Audio Coding

IWSSPA 2025

L. Min, S. Chen, M. Bosi

MuPT: A Generative Symbolic Music Pretrained Transformer

ICLR 2025

X. Qu, ... L. Min, et al. [OpenReview] [Demo]

Whole-song Hierarchical Generation Using Cascaded Diffusion Models

ICLR 2024 (Spotlight)

Z. Wang, L. Min, G. Xia [OpenReview] [Demo]

Polyffusion: A Diffusion Model for Polyphonic Score Generation

ISMIR 2023

L. Min, J. Jiang, G. Xia, J. Zhao [Poster] [Demo]

Teaching and Community Experience

Perceptual Audio Coding, Teaching Assistant at Stanford

Winter 2026

Chuck Development Team, Contributor at Stanford CCRMA

2025 – Present